

Multiwavelength Analysis and Outburst Modeling of X-Ray Binaries

Final Report

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Abstract

Working on this project gave me a hands-on look at how X-ray binary (XRB) systems brighten and fade during outbursts, and why studying them requires piecing together data across different space and ground telescopes. In this report, I present an automated pipeline I built to pull, classify, and model multiwavelength light curves. The pipeline cross-matches 495 targets across X-ray (MAXI, Swift-BAT) and optical (ZTF, ATLAS) archives, using coordinate-matching rules and curated catalogs (WATCHDOG, BlackCAT, XRBcats, HEASARC) to classify each source by compact object (black hole vs. neutron star) and system type (LMXB vs. HMXB). I isolated and fitted 571 outbursts across 125 sources using a Norris pulse convolved with a top-hat kernel to handle instrumental smearing, extracting core shape parameters like rise timescale (τ_1), decay timescale (τ_2), and total duration (T_{90}). Comparing these groups revealed distinct physical patterns: neutron star outbursts brighten much faster than black hole outbursts, while high-mass binaries produce short, tightly clustered bursts compared to the wider range of shapes seen in low-mass binaries. Finally, I simulated the expected optical light from X-rays using a disc-reprocessing model ($F_{\text{opt}} \propto F_X^\beta$). This highlighted clear places where simple models break down, such as high-mass systems where the bright companion star overwhelms the optical light, or systems with unmodelled time delays. Overall, this project helped me bridge the gap between astronomical theory and data engineering, building a practical setup that can scale up for upcoming surveys like LSST.

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1 Introduction

1.1 Classification and Overview

1.1.1 What are X-Ray binaries?

In the 1960s, scientists discovered very bright cosmic X-ray sources with luminosities on the order of about 10^{36} to $10^{38} \text{ erg} \cdot \text{s}^{-1}$; at the time, no physical processes were known to generate such enormous X-ray luminosities. This was later explained by the discovery of X-ray binaries. The intense X-rays from these sources were recognised to arise from the accretion of matter from a companion star onto the surface of a compact star. The companion star is usually a normal star or a white dwarf transferring mass onto the compact object, which might be a neutron star or a black hole. It accretes matter through either Roche Lobe Transfer or Stellar Wind accretion. The matter accreted from the companion star flows onto the compact star under the influence of its gravity. As the inflowing matter gains angular momentum, it spirals onto the compact star, forming an accretion disk. The violent collisions between particles heat the gas to very high temperatures. In this process gravitational energy is converted to kinetic energy and eventually into radiation in the form of X-rays. There are two types of X-ray binaries:

- **Low Mass X-Ray Binaries:** Low-Mass X-ray Binaries (LMXBs) are interacting stellar systems where a compact object (a neutron star or black hole) strips matter from a low-mass companion star ($\leq 2 M_{\odot}$) via Roche-lobe overflow. This mass transfer is driven by gravitational radiation or magnetic braking in tight orbits (≤ 2 days), or by nuclear stellar evolution in wider systems. The resulting emission is strongly dominated by high-energy X-rays emitted from the inner accretion disk, which thermally reprocess into optical light at the outer disk boundaries. Because of their long operational lifetimes ($\sim 10^7 - 10^9$ years), LMXBs are primarily distributed among older stellar populations toward the galactic center and within globular clusters, and their sustained, prolonged accretion typically suppresses the magnetic fields of resident neutron stars, preventing periodic pulsar pulsations.
- **High Mass X-Ray Binaries:** High-Mass X-ray Binaries (HMXBs) are young, short-lived stellar systems ($\sim 10^5 - 10^7$ years) where a compact object accretes matter from a massive companion star ($\geq 10 M_{\odot}$) primarily via strong, line-driven stellar winds. Because the companion does not necessarily fill its Roche lobe, the system's total emission is optically dominated by the luminous companion star, while the X-ray generation is governed by the velocity and density of the wind intercepting the compact remnant. Due to their brief evolutionary timescales, HMXBs are concentrated almost exclusively along the galactic plane alongside active star-forming regions. Furthermore, because their accretion phases are relatively short, the strong natal magnetic fields of resident neutron stars are preserved rather than buried, causing them to frequently manifest as highly periodic, accretion-powered X-ray pulsars.

2 The Physics of Accretion

2.1 Disk Dynamics and Instabilities

To fall inward toward a compact object, gas in an accretion disk must find a way to slow down and lose its orbital energy, and the molecular viscosity was way too weak to account for it. This mystery was solved in 1991 by theorists Steven Balbus and John Hawley through the Magnetorotational Instability (MRI) (Balbus & Hawley, 1991).

Because the accretion disk is made of plasma (charged particles), it physically traps the background magnetic field lines running through it. The inner lanes of the disk orbit much faster than the outer lanes. As these inner particles race ahead, they drag and stretch the magnetic field lines behind them like tense rubber bands. This stretching creates a backward pulling force (a magnetic brake) on the fast inner gas. By slowing the inner gas down, it loses its speed, can no longer fight gravity, and falls inward toward the central object. This continuous tug-of-war eventually leads to full-blown magnetic turbulence, creating the massive friction needed to continuously drive matter down into X-ray binaries.

Another question that arose during the research on X-ray binaries was why certain X-ray binaries look incredibly dim despite constantly consuming matter. This was explained by Advection Dominated Accretion Flows. When accretion rates drop to a low, the density in the accretion disk becomes too low for the heat to be radiated out due to collisions between the particles. What ends up happening is that the heat is advected into the event horizon of the black hole. Without radiative cooling, the internal gas

pressure causes the disk to puff up into a geometrically thick, donut-like shape. This mechanism perfectly explains why certain X-ray binaries in quiescent states appear remarkably dim despite ongoing accretion.

2.2 The Magnetospheric Boundary

At the inner edge of the accretion disk, the behavior of the infalling plasma is entirely dictated by the Magnetospheric Boundary, which is defined by the Alfvén radius (R_A). This is the physical boundary where magnetic effects dominate. For a standard neutron star with a magnetic field of 10^{12} G accreting near the Eddington limit, this boundary truncates the thin disk at roughly 100 stellar radii. Once the matter hits the Alfvén radius, its fate is decided by a strict centrifugal tug-of-war. If the neutron star’s spin frequency is slower than the orbital frequency of the disk gas, the material is forced to corotate with the star and is channeled along magnetic field lines into narrow accretion columns hitting the poles, producing an accretion-powered pulsar. Conversely, if the star spins faster than the local disk material, it creates a centrifugal barrier (the propeller phase) that rejects the gas, likely casting it out as a wind. Over time, these opposing torques drive the system toward a critical equilibrium spin frequency—typically a few tenths of a Hertz—where the net torque on the compact object is zero.

3 Observational Instrumentation

3.1 Grazing Incidence Optics

Unlike traditional optical telescopes that use front-facing mirrors, X-ray telescopes must utilize grazing incidence optics because high-energy photons will penetrate or be absorbed by a mirror rather than reflecting off it. To achieve efficient reflection, incoming X-rays must strike the reflecting surface at a shallow grazing angle smaller than a critical threshold, which is typically on the order of just one degree (0.017 radians). The heart of the Chandra X-Ray Observatory features a highly precise configuration known as the Wolter-I geometry. This configuration forces incoming photons to undergo two successive reflections to provide exceptional imaging across a wide field of view: first striking a paraboloid of revolution mirror, followed immediately by a hyperboloid of revolution mirror. To maximize the total photon collecting area while keeping the telescope compact, Chandra features multiple pairs of these concentric, shallow angle mirror shells nested tightly inside one another, all engineered to share the exact same focal plane. This design improved angular resolution by an order of magnitude over its predecessors, allowing scientists to observe various phenomena at a finer scale.

3.2 Advanced Spatial Resolution via Occultation

To overcome the fundamental diffraction limits of traditional grazing incidence mirrors, the proposed X-ray Big Occulting Steerable Satellite (X-BOSS) framework replaces complex spatial imaging with a timing solution. Rather than allowing the entire photon flux to flood the detector simultaneously, which blinds the telescope by causing bright central accretion zones to completely wash out adjacent, subtler structures, X-BOSS utilizes a maneuverable artificial satellite positioned thousands of kilometers ahead of a primary observatory like *Chandra*.

As the sharp physical edge of this occulting satellite slowly cuts across the line of sight, it blocks incoming photons slice by slice. Because modern X-ray detectors can record individual photon arrival times down to microsecond precision, the rate at which the photon counts plummet or rise acts as a direct mathematical map of the target’s geometry. By tracking these sharp drops across multiple sweep angles, some algorithms can reconstruct the brightness profiles of close orbit binary environments and inner accretion flows at microarcsecond scales, achieving a resolution that would otherwise require a mirror array the size of a football stadium.

4 Literature Review: Source State, Jets, and Multiwavelength Coupling

Alongside building the pipeline, I read four papers that each shaped a specific design choice below: how burst/outburst properties should be interpreted relative to accretion state, why jet ejections happen when they do, and why the optical reprocessing model in Section 8 takes the form it does.

4.1 X-ray Bursts and Accretion States in Atoll Sources

Scientists studied 4U/MXB 1636–53, a low-mass X-ray binary system containing a neutron star. These types of systems are called “atoll sources” because their X-ray colors trace a specific path on a graph that looks like an island chain. As the star pulls in material from its companion (a process called accretion), it moves between two main states:

- **Island state:** The source is faint, emits hard X-rays, and changes brightness slowly.
- **Banana state:** The source gets brighter, emits softer X-rays, and changes brightness rapidly.

Scientists knew that a source’s state acts as an indirect indicator of how fast it is pulling in material. By examining 61 thermonuclear X-ray explosions (Type I bursts) from archival space telescope data, researchers checked the length and peak temperature of each burst against the source’s state at the time.

They found a strong pattern: bursts during the island state are longer and hotter than bursts during the banana state. Because these explosions only happen on the solid surface of a neutron star and depend on the layer of fuel built up on top, this link proved that burst length and temperature directly track the star’s material intake rate.

The bigger surprise came from looking at the regular, steady X-ray brightness of the star. Overall brightness does not track the material intake rate nearly as cleanly. When comparing two time periods where the star was in the exact same state, its overall brightness could be completely different. This proved that overall brightness alone is a bad way to estimate how fast a star is consuming material. Looking at the overall state (color and rapid brightness changes) is far more reliable.

4.2 A Unified Model for Black Hole Jets

Researchers compiled radio and X-ray data for four black hole systems during major radio outbursts to figure out exactly which X-ray state triggers powerful plasma jets.

Here is how the process works:

1. **Low/Hard State:** The black hole launches a steady, compact, slow-moving jet.
2. **Intermediate State:** As the outburst grows and the X-ray spectrum softens, the steady jet stays active and maintains its connection to the X-ray energy.
3. **The Bright Radio Flare:** Right near the end of this transition phase, a powerful, fast-moving jet is suddenly launched. This new, high-speed jet crashes into the slower material released earlier, creating a powerful shockwave that produces a bright radio flare.
4. **High/Soft State:** The jet shuts off completely as the system settles into this state.

This pattern allowed scientists to map a “jet line” on a graph of energy versus brightness. Crossing this boundary from right to left marks the exact moment the fast jet is launched and the main radio flare occurs.

This behavior also connects directly to the inner disk of material swirling around the black hole. In the hard state, the inner edge of this disk stays far away from the black hole. As the intake rate rises, the disk pushes inward toward the black hole, and jet speeds skyrocket right before the disk reaches its inner limit.

This explains why an X-ray light curve on its own cannot tell us when a jet was launched—detecting a jet requires simultaneous radio tracking. However, it tells us that we should treat the rising phase of an outburst differently from the fading phase. The optical and infrared light from a jet is strongest during the rise phase and shuts down once the black hole transitions into its soft state.

4.3 Disc and Jet Contributions to the Optical/Infrared Emission

In optical and near-infrared light, three main mechanisms can drive emission: X-ray reprocessing in the disc, heat from friction inside the disc, or direct radiation from the compact jet. Whether the dominant mechanism depends on the type of compact object (black hole versus neutron star) was examined using observational data.

By gathering optical, near-infrared, and X-ray data from 33 X-ray binary systems, tight mathematical links were found in the hard state:

- **Black hole systems:** Light levels in the optical and infrared follow a strict pattern relative to X-ray brightness across eight orders of magnitude.
- **Neutron star systems:** Follow a nearly identical pattern over seven orders of magnitude.

In the soft state, near-infrared emission from black hole systems drops sharply compared to the hard state, while optical emission drops only slightly. This indicates that the jet extends out to near-infrared wavelengths and gets shut off in the soft state, just like the radio jet. The optical light, on the other hand, remains mostly driven by disc reprocessing.

Comparing theoretical models against the data showed clear differences between source types:

- **For black holes:** Both the disc-reprocessing model and the jet model match the observed data well, while heat from friction inside the disc alone does not.
- **For neutron stars:** The optical data is far better explained by disc reprocessing alone. The jet model cannot reproduce the pattern because neutron stars process radiation efficiently and rarely reach jet-dominated states.

Why this matters to the project: This result directly justifies the formula used in our optical-reprocessing simulation. It allows us to set different scaling values for black hole and neutron star samples separately, rather than assuming a single textbook value for every object regardless of its type.

4.4 A Long-Term Optical X-ray Correlation in a Single Source

Applying reprocessing principles to a single system, researchers examined 4U 1957+11, a persistently active low-mass X-ray binary with a historically soft X-ray spectrum and no confirmed thermonuclear bursts. Using three years of optical photometry cross-correlated against soft X-ray monitoring data, a significant long-term optical–X-ray correlation was identified with a best-fit slope of $\beta \approx 0.5$ ($F_{\text{opt}} \propto F_X^\beta$). This matches the theoretical value expected for simple disc reprocessing, while the measured time lag indicates prompt reprocessing on the disc surface rather than a delayed viscous effect.

Comparing the position of 4U 1957+11 on an optical–X-ray luminosity diagram against established black hole and neutron star samples provides a diagnostic for the compact object’s nature:

- **At standard distance estimates (7–10 kpc):** The system’s luminosity aligns strictly with neutron star sources.
- **To match soft-state black holes:** The source would need to reside beyond 20 kpc.

This provides an independent method for constraining compact object types using X-ray and optical data when a dynamical mass measurement is unavailable.

4.5 Synthesis

All of these features help us ingrain these features in our code:

1. **State over flux:** Outburst state and profile shape provide far more reliable diagnostics than raw flux levels.
2. **Component distinction:** X-ray light curves and jet behavior do not always align, making it essential to clearly distinguish between disc-reprocessing and jet-synchrotron contributions when modeling X-ray and optical data.
3. **Empirical slope fitting:** The reprocessing index β varies across populations rather than acting as a universal constant, justifying our approach of fitting β separately for black hole and neutron star samples instead of assuming a default value of 0.5.

5 Multiwavelength Data Extraction Pipeline

5.1 Cross-Matching Pipeline Architecture

The core challenge both pipelines solve is finding and automating the extraction of X-ray and optical band data for X-ray binaries. To automate this, the first code approaches the problem by either tapping directly

into database APIs or cleverly exploiting HTML web URL patterns to hunt down files. It successfully crossmatches each target across the MAXI and Swift space telescopes for X-ray data, and pairs them with ZTF for optical data, while automatically routing the request to the ATLAS telescope as a fallback if ZTF has no coverage. While both pipelines effectively perform the exact same core matching and fallback tasks, the second code introduces crucial upgrades: the script is written much cleaner, is easier to maintain, and switches its baseline reference to a dedicated X-ray binary master catalog (`xrbc`) instead of a general MAXI catalog, ensuring a far more accurate target list from the very start. By cleanly organizing these data streams into uniform local spreadsheets, the pipeline sets us up for the next step: using this data to generate light curve plots that visually map out these outbursts.

5.2 Automated Outburst Detection

Before fitting an outburst, it must first be identified within irregular light curve data across hundreds of sources. While automated thresholding methods (such as point-level hysteresis) can scan for candidates, interactive manual selection was determined to be the most reliable method for defining precise event bounds.

To ensure consistency and accuracy, interactive span selection was used to mark outburst start and end times across sources, as implemented in `fred.spanned.py`. Once marked, these selected time windows are saved and loaded directly into the pipeline to isolate individual events. Each candidate span is verified to contain sufficient data points and cleared above the quiescent baseline before being passed into the model-fitting stage.

5.3 Outburst Profiles

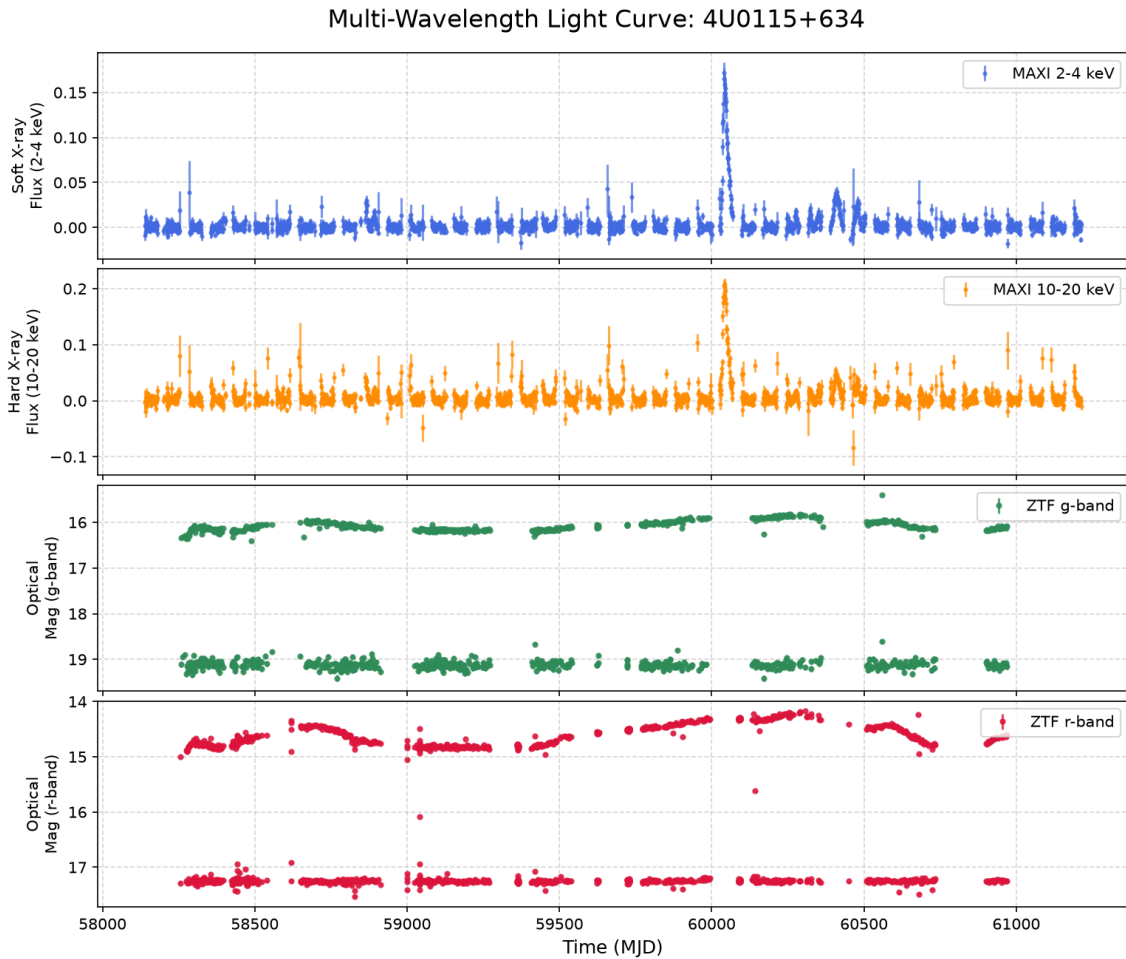


Figure 1: Multiwavelength outburst profile for 4U 0115+634 (MAXI X-ray bands + ZTF/ATLAS optical), showing the detected outburst window. *[Placeholder – regenerate from the current Total_Data run.]*

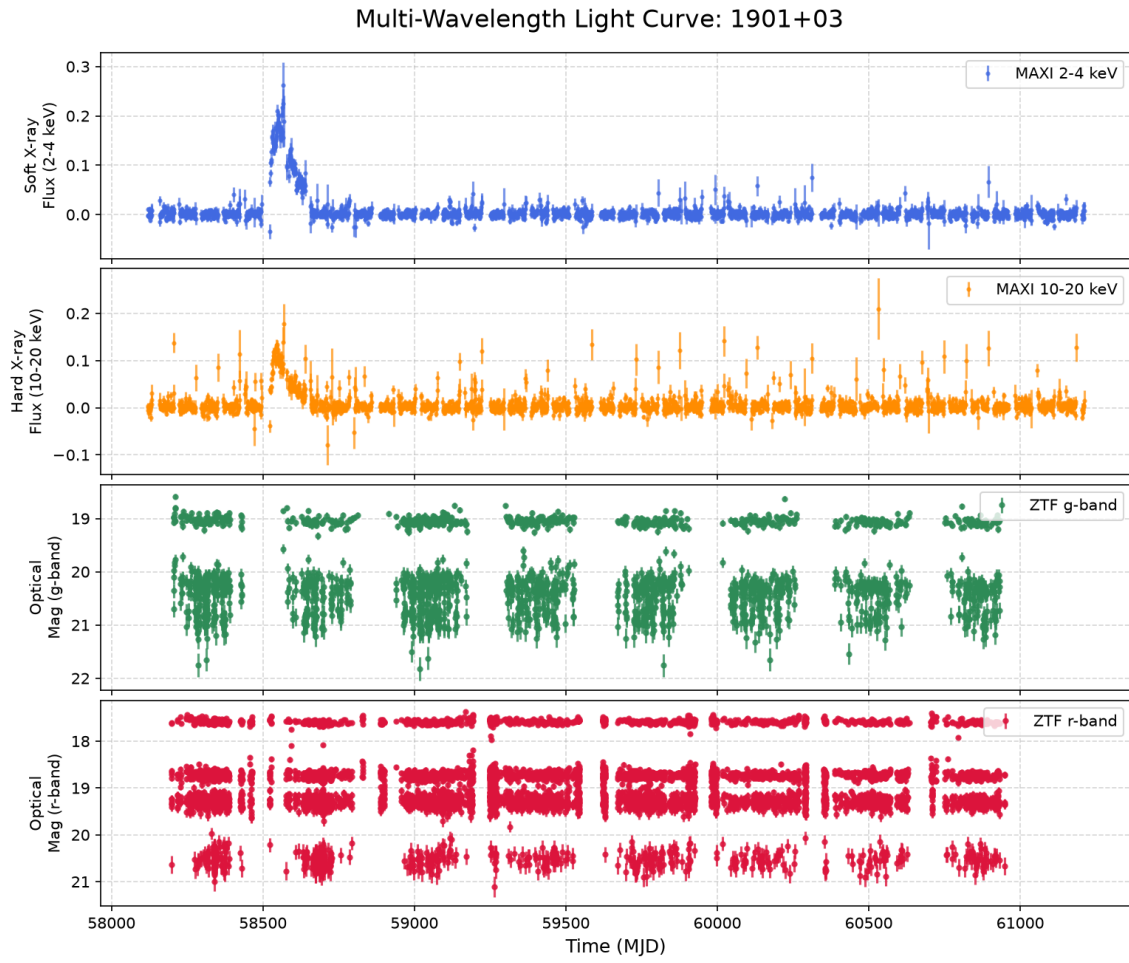


Figure 2: Multiwavelength outburst profile for 1901+03. *[Placeholder – regenerate from the current Total_Data run.]*

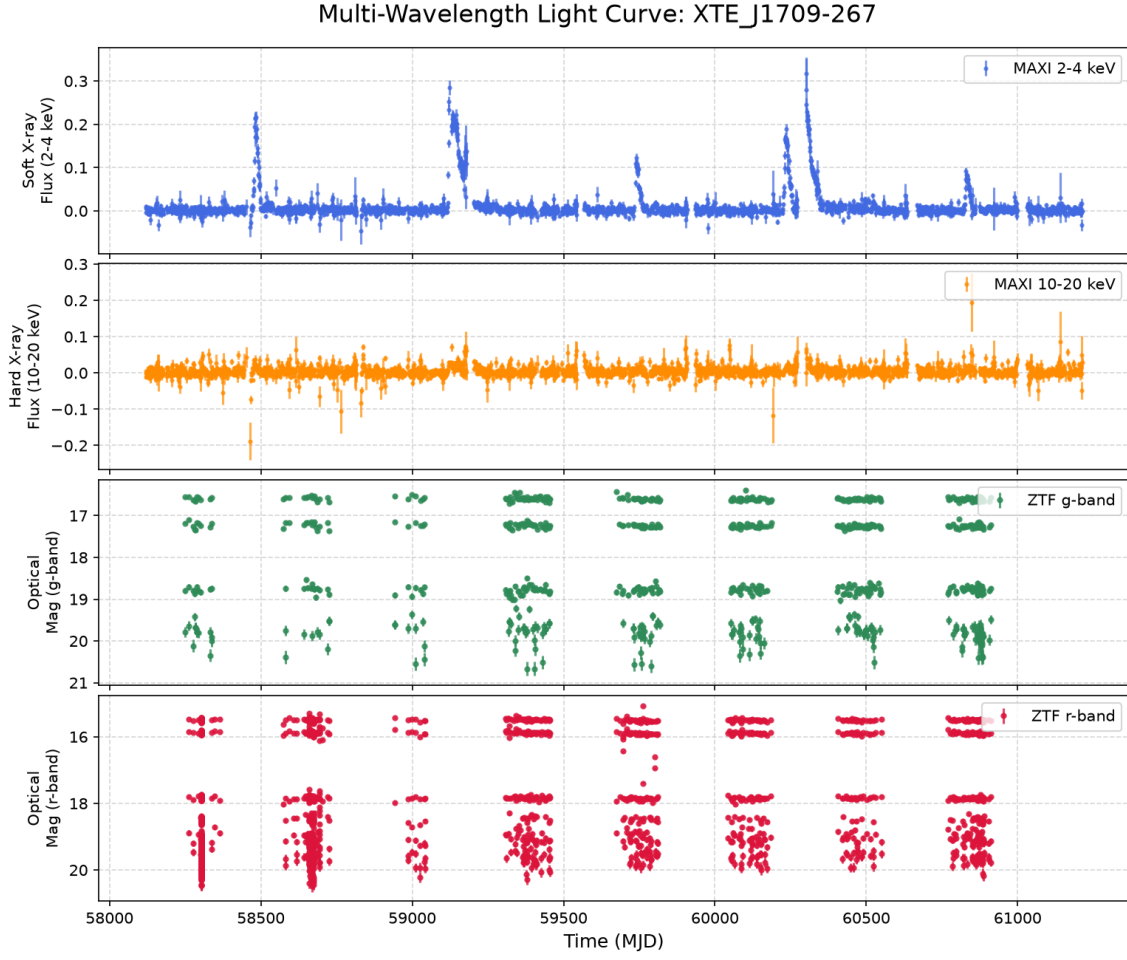


Figure 3: Multiwavelength outburst profile for XTE J1709–267. *[Placeholder – regenerate from the current Total_Data run.]*

6 Source Classification Pipeline

Every source has to be tagged with a compact-object type (BH/NS) and a system type (LMXB/HMXB) before the population comparisons or the reprocessing fit happen to be meaningful. `Classifier.py` implements this as a hierarchical cross-match against curated catalogs and SIMBAD.

The original design queried VizieR/HEASARC once per source, in a simple loop, but slowly (minutes, not seconds, for a few hundred sources) and at constant risk of tripping SIMBAD's rate limiter. The version documented here instead downloads each reference catalog once, caches it to disk as an Astropy table, and then cross-matches the entire source list against it locally with `astropy.coordinates.SkyCoord` cone-matching (10'' radius by default). SIMBAD is only contacted for (a) name coordinate resolution when a filename gives a name rather than coordinates, and (b) the final fallback for sources that match none of the curated catalogs. After the first run, the whole pipeline can run fully offline against the cache.

The hierarchical decision tree is:

1. **BH filter** – WATCHDOG (J/ApJS/222/15) and BlackCAT (J/A+A/587/A61), essentially a complete census of confirmed BH LMXBs.
2. **LMXB filter** – the XRBcats Galactic LMXB catalog (J/A+A/675/A199).
3. **HMXB filter** – the HEASARC `hmxbcatalog`.
4. **SIMBAD fallback** – object-type (`otype`) lookup, used only as weak evidence when nothing above matches.

6.1 Coordinate Resolution

To resolve non-coordinate query names, target positions are determined using a prioritized resolution cascade:

1. **Direct Parsing** (`_resolve_direct`): IAU J-names (e.g., MAXI J2303+088) are parsed directly. If implied positional precision falls within the match radius, coordinates are accepted outright; otherwise, they are cross-checked against curated catalogs at a 600'' sanity radius to adopt the precise catalog position. Non-J catalog names (4U, 1A, etc.) are similarly verified against curated catalogs across candidate declination precisions.
2. **Exact SIMBAD Match**: Queries SIMBAD using the target name as provided.
3. **SIMBAD Cone Search**: Executes a single-indexed coordinate cone search around a coarse local position hint when available.
4. **SIMBAD Alias Matching**: Retries resolution using generated alias variants (e.g., trying 4U 0535+26 if 4U 0535+262 fails).
5. **SIMBAD Wildcard Search**: Performs a trailing-wildcard query as a final fallback, accepted only if it resolves unambiguously to a single object.

6.2 Compact-Object and System-Type Disambiguation

Because curated reference catalogs lack universal compact-object indicators, system classifications are inferred directly from diagnostic column values:

- **LMXB Catalog Matches**: Coherent pulsations or millisecond pulsar flags imply a neutron star (NS) surface. Thermonuclear burst, atoll, or Z-source descriptions in the `XrayType` field likewise indicate an NS. Confirmed LMXBs lacking explicit BH catalog hits or NS diagnostic flags default to NS (flagged for manual verification).
- **HMXB Catalog Matches**: Populated `pulse_per` values identify accreting NS pulsars, while an explicit "BH" flag in the `class` column indicates a black hole HMXB.
- **BH Matches**: Direct matches to WATCHDOG or BlackCAT authoritatively assign `compact_object = BH`.

System type (LMXB vs. HMXB) is resolved by choosing the closer positional match between the corresponding catalogs. If only a BH catalog match exists, its internal `Type/Class` column is queried first, falling back to an LMXB default if unpopulated.

6.3 Classification Results

Processing the 495 sources in `TotalData` generated `classification_results.csv` (summarized in Table 1). High confidence matches were obtained for 174 sources (35%), 8 matched with medium confidence, and 313 (63%) defaulted to low-confidence SIMBAD fallback or remained unresolved—highlighting the incompleteness of curated catalog coverage relative to archival MAXI/ZTF source lists.

Table 1: Summary of the 495-source classification run (`classification_results.csv`).

Matched Catalog	Count	Fraction
SIMBAD (fallback)	304	61.4%
XRBCats LMXB (<code>lmbx_xrbcats</code>)	122	24.6%
HEASARC <code>hmbcat</code>	54	10.9%
WATCHDOG (<code>watchdog_bhxb</code>)	6	1.2%
None (unresolved)	9	1.8%
Confidence: High / Medium / Low	174 / 8 / 313	35.2% / 1.6% / 63.2%
Compact Object: NS / BH / Unknown	132 / 44 / 319	26.7% / 8.9% / 64.4%
System Type: LMXB / HMXB / Unknown	131 / 58 / 306	26.5% / 11.7% / 61.8%

7 Outburst Model Fitting

7.1 From Dual-Model Comparison to a Unified Norris-Tophat Model

The midterm version of this pipeline fit two competing empirical outburst shapes to each light curve and used the Akaike Information Criterion to pick a winner: the Fast-Rise Exponential-Decay (FRED) profile,

$$F(t) = A \exp\left(-\frac{t-t_0}{\tau_d}\right) \left(1 - \exp\left(-\frac{t-t_0}{\tau_r}\right)\right),$$

and the Norris profile, more commonly used for high-energy transients,

$$F(t) = A \exp\left(-\frac{\tau_1}{t-t_0} - \frac{t-t_0}{\tau_2}\right).$$

In the current implementation, this comparison has been simplified into a single model: the bare Norris pulse above, convolved with a normalized boxcar (tophat) kernel of width W ,

$$F_{\text{smeared}}(t) = (F_{\text{Norris}} * \text{tophat}_W)(t),$$

evaluated via FFT convolution on a fixed, fine, uniform MJD grid built once per fit. The convolution kernel lets a single sharp Norris pulse absorb real, physical smearing from finite MAXI binning and instrumental averaging without needing a second competing model, so W is fit as a fifth free parameter alongside A , t_0 , τ_1 (rise), and τ_2 (decay), rather than being fixed at the pixel/bin scale.

Outburst spans are read directly from the pre-built outburst catalog produced by the detector, filtered to the X-ray/MAXI, 2-20 keV rows with at least 25 photometric points, so the same objectively-detected span is reused consistently across all four MAXI energy bands (2-4, 4-10, 10-20, and the combined 2-20 keV) for a given outburst.

7.2 Shape-Parameter Extraction and Goodness of Fit

Each fit is performed with bounded, weighted nonlinear least squares, and its quality is summarised by the reduced χ^2 and a small-sample corrected Akaike Information Criterion,

$$\text{AIC} = \chi^2 + 2k + \frac{2k(k+1)}{n-k-1},$$

for k free parameters and n data points, retained now purely as a per-fit diagnostic rather than for model selection. From the fitted, convolved curve, five shape parameters are read off numerically over the

outburst window: the rise timescale τ_1 , decay timescale τ_2 , time-to-peak (relative to t_0), and T_{90} —the interval containing the central 90% of the fitted fluence, computed from the cumulative trapezoidal integral of the model curve. These four quantities are what feed the population histograms below, deliberately in preference to peak flux or persistent intensity.

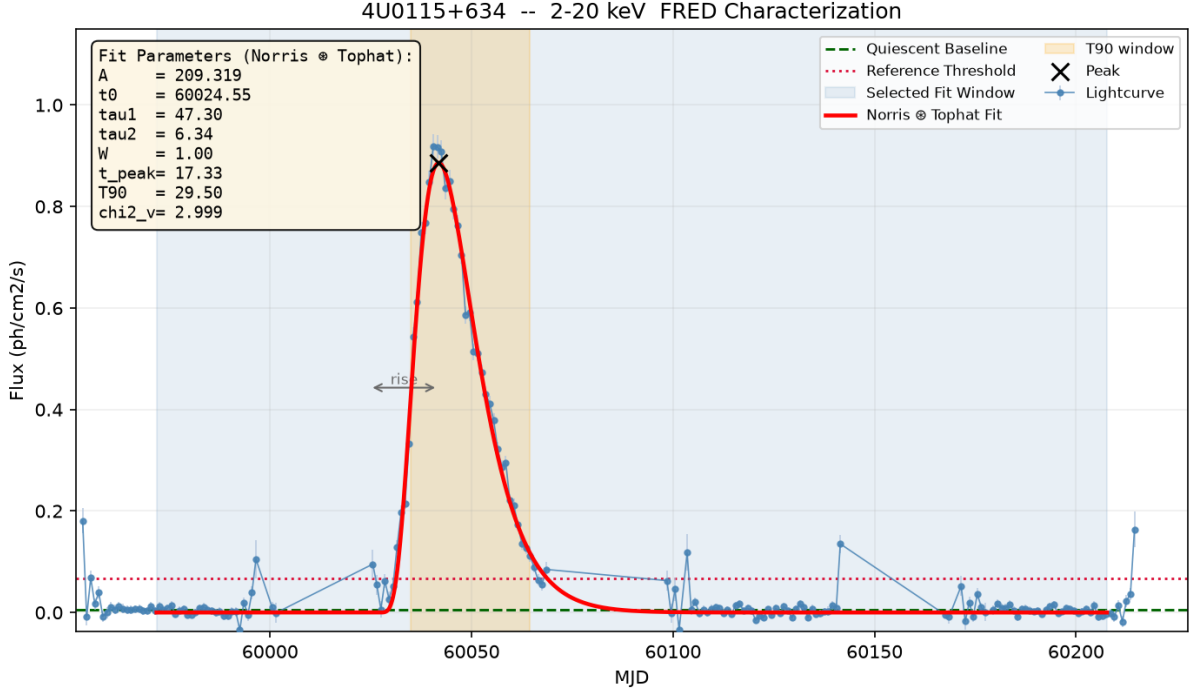


Figure 4: Norris-tophat fit overlaid on the 2–20 keV MAXI light curve of 4U 0115+634.

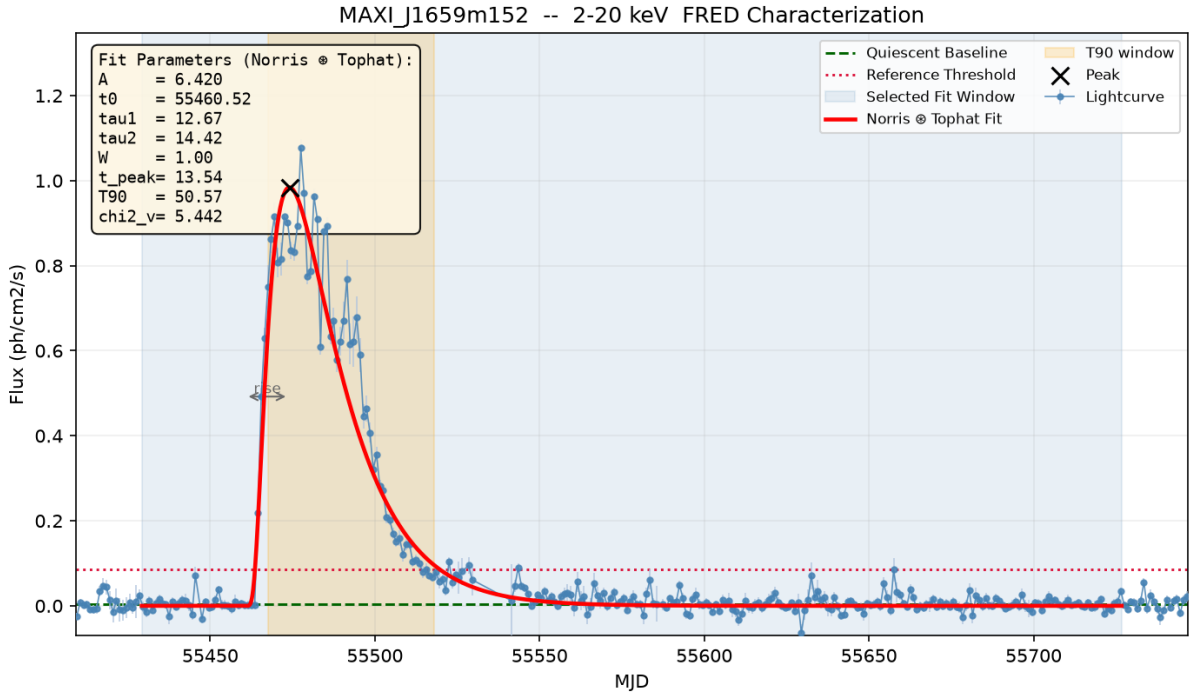


Figure 5: Norris-tophat fit overlaid on the 2–20 keV MAXI light curve of MAXI_J1659m152.

7.3 Population Histograms by Compact Object and System Type

The analysis evaluates 571 fitted outbursts across 125 unique X-ray sources in the 2–20 keV energy band. Outburst profiles are characterized by four shape parameters: rise timescale (τ_1), decay timescale (τ_2), time-to-peak ($t_{\text{peak}} - t_0$), and total duration (T_{90}). Sources are categorized by compact object (Black Hole vs. Neutron Star) and system type (Low-Mass X-ray Binary vs. High-Mass X-ray Binary). Unclassified sources are excluded from the grouped histograms to maintain population clarity.

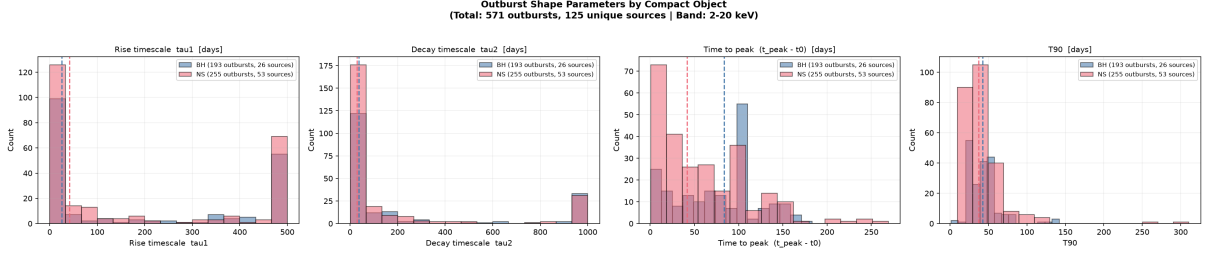


Figure 6: Outburst shape parameters (rise timescale, decay timescale, time-to-peak, T_{90}) grouped by compact object, 2–20 keV band. 571 outbursts, 125 unique sources.

Comparing Black Holes (193 outbursts, 26 sources) and Neutron Stars (255 outbursts, 53 sources) shows that both populations are dominated by short-duration outbursts with rapid rise and decay timescales (< 50 days). However, their rise behaviors differ noticeably: Neutron Star outbursts peak rapidly (mostly within 50 days), whereas Black Hole outbursts take longer to reach maximum brightness, typically peaking between 80 and 110 days. Decay rates and overall durations are broadly similar across both groups, though both exhibit a minor sub-population with extended rise timescales ($\tau_1 \sim 500$ days).

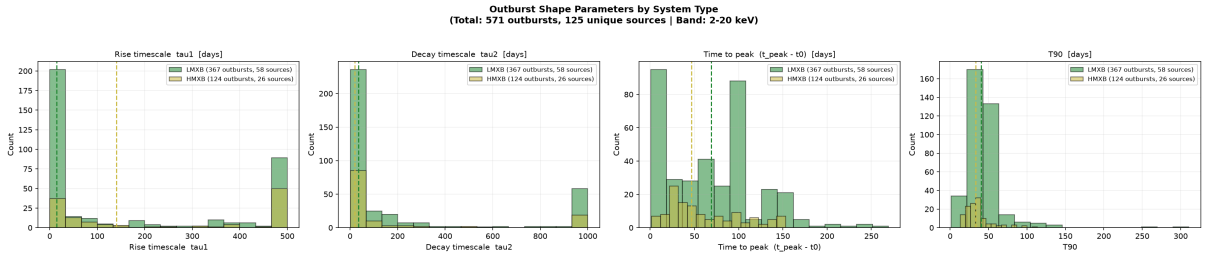


Figure 7: Outburst shape parameters grouped by system type (LMXB/HMXB), 2–20 keV band. Same underlying sample as Figure 6.

Grouping by system type reveals a sharp contrast between Low-Mass X-ray Binaries (367 outbursts, 58 sources) and High-Mass X-ray Binaries (124 outbursts, 26 sources). HMXBs display a tight, uniform distribution characterized by short durations, fast rise/decay times, and quick peaks (< 50 days). In contrast, LMXBs show much broader parameter distributions, accounting for almost all long-duration events as well as the extended rise-time tail (~ 500 days).

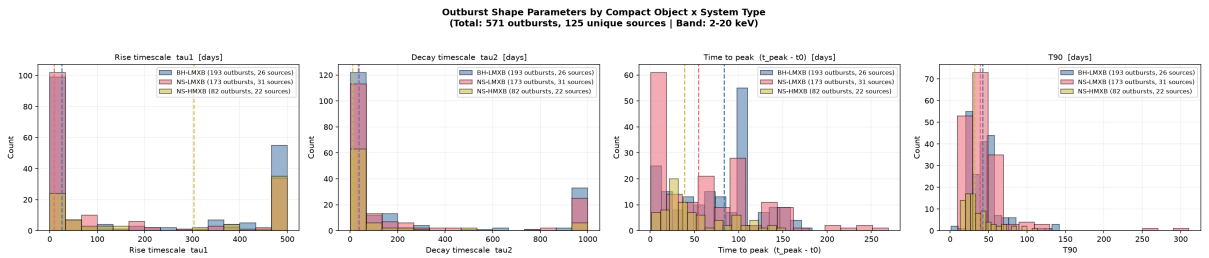


Figure 8: Combined breakdown by compact object $\times$ system type (BH-LMXB, NS-LMXB, NS-HMXB).

The combined breakdown (BH-LMXB, NS-LMXB, and NS-HMXB) highlights how system geometry shapes outburst profiles. NS-HMXBs are almost exclusively fast, short-duration events, while BH-LMXBs

and NS-LMXBs drive the broader time-to-peak and duration distributions. Furthermore, the secondary population of slow-rising outbursts ($\tau_1 \sim 500$ days) is isolated strictly to LMXBs, highlighting the need for dedicated follow-up to determine whether this tail reflects slow accretion dynamics or model-fitting limitations on long baselines.

8 Optical Reprocessing and Forward Simulation

8.1 Model Formulation and Methodology

To simulate optical counterparts from X-ray flux, we adopt the classic disc-reprocessing power law

$$F_{\text{opt}} \propto F_X^\beta \quad (1)$$

evaluated as a dimensionless shape function ($F_{\text{opt}} = F_X^\beta$). To avoid parameter trade-offs that created noise artifacts in earlier iterations, the model incorporates several key design refinements:

- **Fixed-Effects Slope Fitting:** We estimate β separately for BH and NS sources using a within-source (mean-centered) fit on log-log flux pairs. This ensures that differences in distance or extinction across sources do not bias the estimated slope.
- **Rise-Phase Isolation:** Fitting is restricted exclusively to the rise phase of each outburst, where the optical emission is dominated by hard-state disc reprocessing rather than complex soft-state decays.
- **Quiescent Excess Subtraction:** Each source’s 10th-percentile optical flux is subtracted as a quiescent baseline, ensuring we fit the reprocessed disc excess rather than constant donor-star light.
- **Noise-Floor Tapering:** Points with MAXI SNR < 3 are weighted using a smooth logistic taper (width 1.5 in SNR) to prevent artificial step-function jumps near the detection boundary.
- **Fallback Thresholds:** If a population yields fewer than 5 valid data pairs or fits an extreme slope ($|\beta| > 3$), the pipeline falls back to the theoretical standard $\beta = 0.5$.

8.2 Empirical Fit Results

Running the fit across the sample yielded $\beta_{\text{NS}} = 0.029$ for neutron stars and triggered the $\beta_{\text{BH}} = 0.500$ fallback for black holes.

The BH population failed to accumulate enough quasi-simultaneous rise-phase data pairs, causing the code to default to $\beta = 0.500$. Figures 9 and 10 show this fallback model applied to MAXI J1631–478. Figures 11 and 12 highlight MAXI J1634–473, where the observed optical light peaks 80–100 days after the X-ray peak. This physical time lag cannot be captured by an instantaneous model.

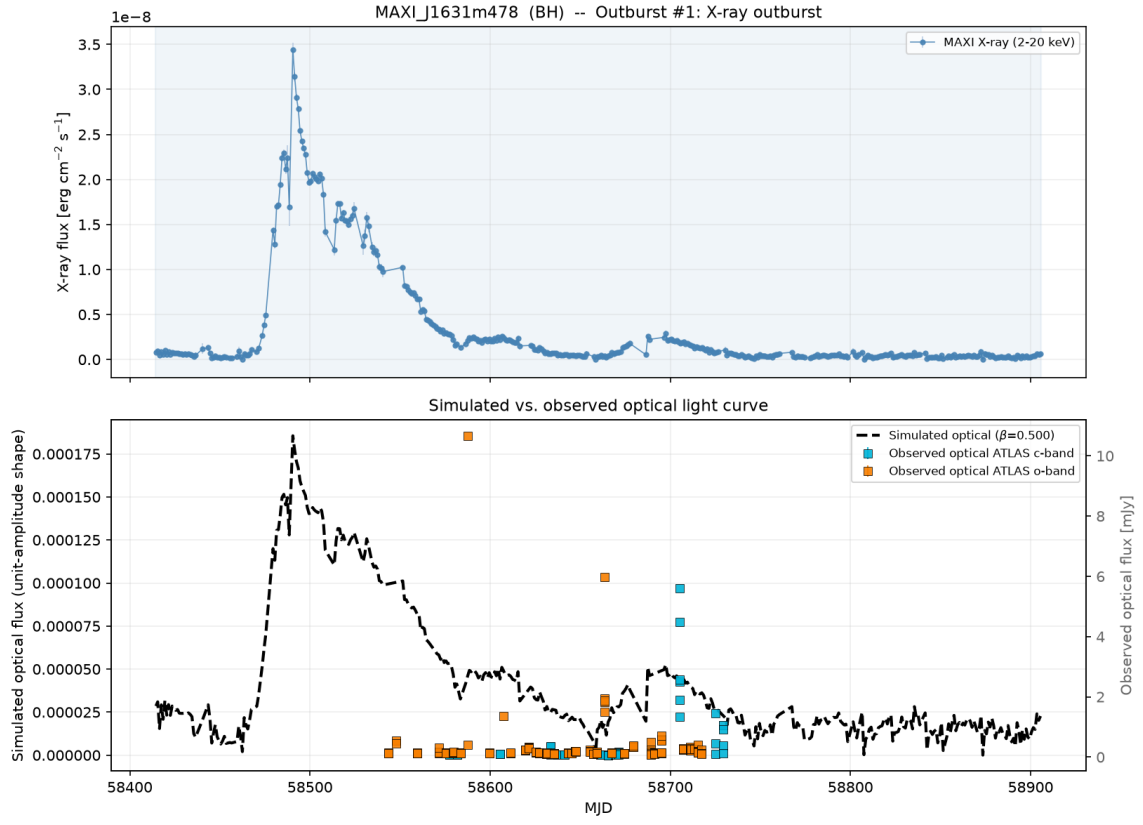


Figure 9: MAXI J1631–478, outburst #1. Simulated curve uses the default $\beta = 0.5$ fallback due to insufficient rise-phase pairs.

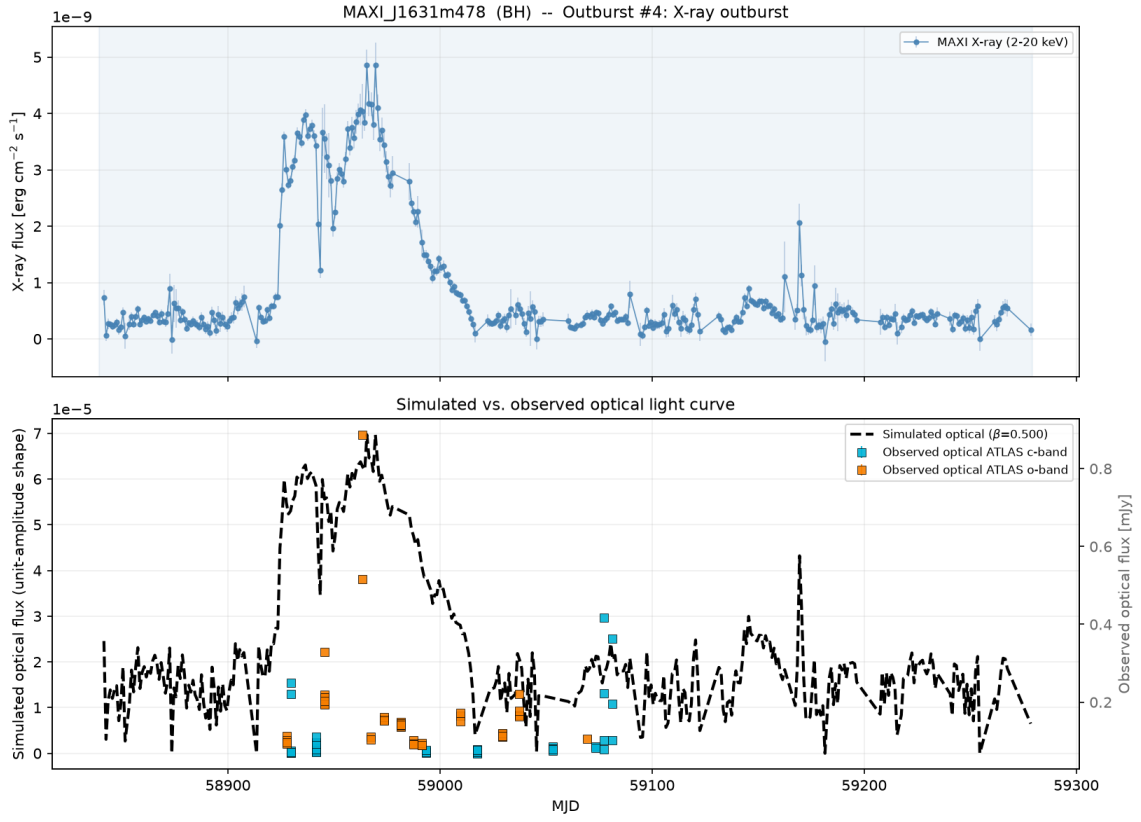


Figure 10: MAXI J1631–478, outburst #4. A rare instance where the simulated X-ray profile and observed optical points roughly align.

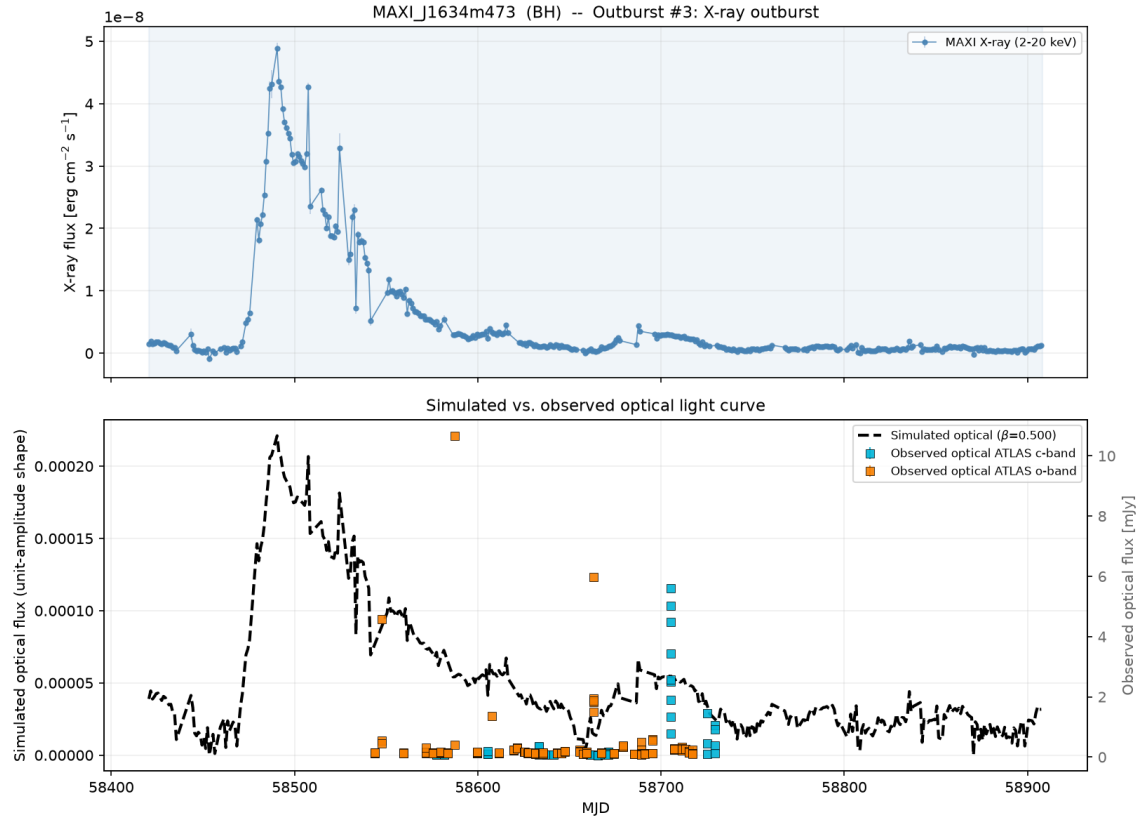


Figure 11: MAXI J1634–473, outburst #3. The observed optical light peaks ~ 80 – 100 days after the X-ray peak.

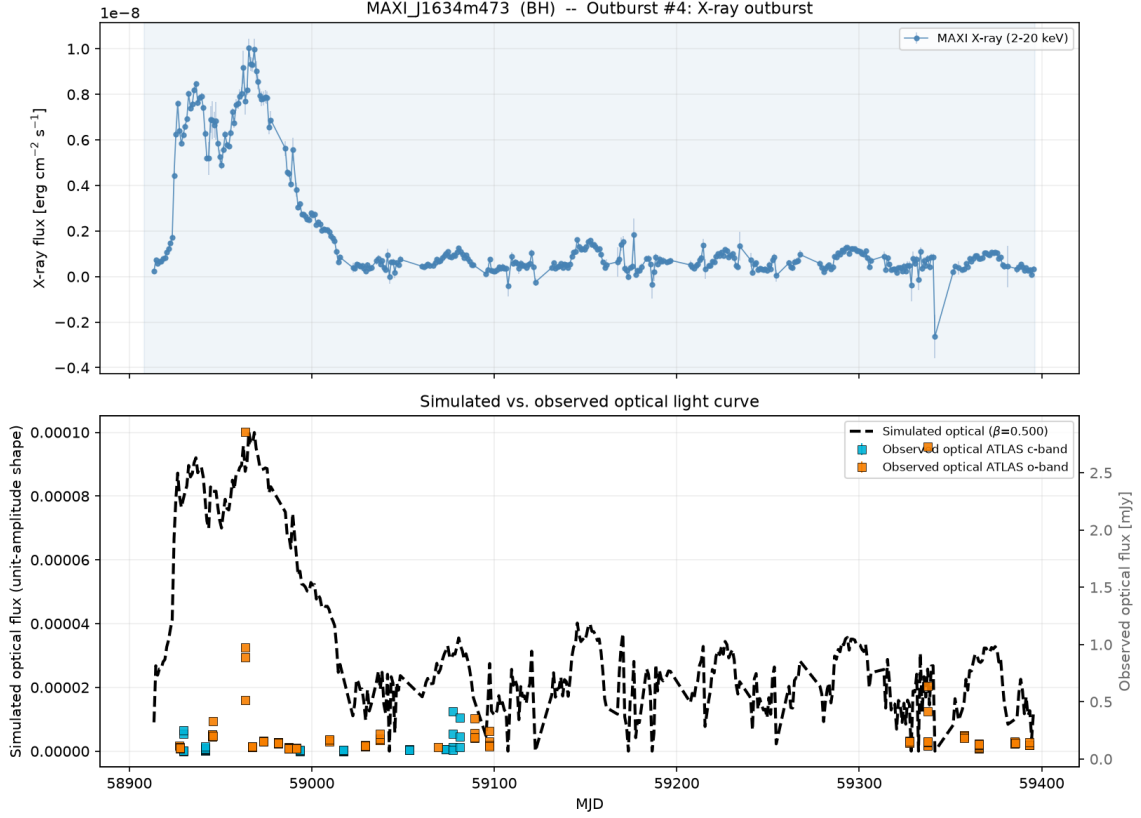


Figure 12: MAXI J1634–473, outburst #4. Optical emission lags behind the X-ray decay, indicating an unmodelled delay.

8.3 Diagnostics and Failure Modes

Although the NS class met the minimum pair threshold, $\beta_{\text{NS}} = 0.029$ is unphysically flat. As $\beta \rightarrow 0$, $x^\beta \rightarrow 1$ for any positive x , turning the model into an on/off step function. As shown in Figures 13 and 14, low-level X-ray noise above zero gets mapped to near-unit optical amplitude, producing an artificial sawtooth pattern. The current upper ceiling ($|\beta| \leq 3$) failed to reject this unphysical fit because it lacks a lower bound.

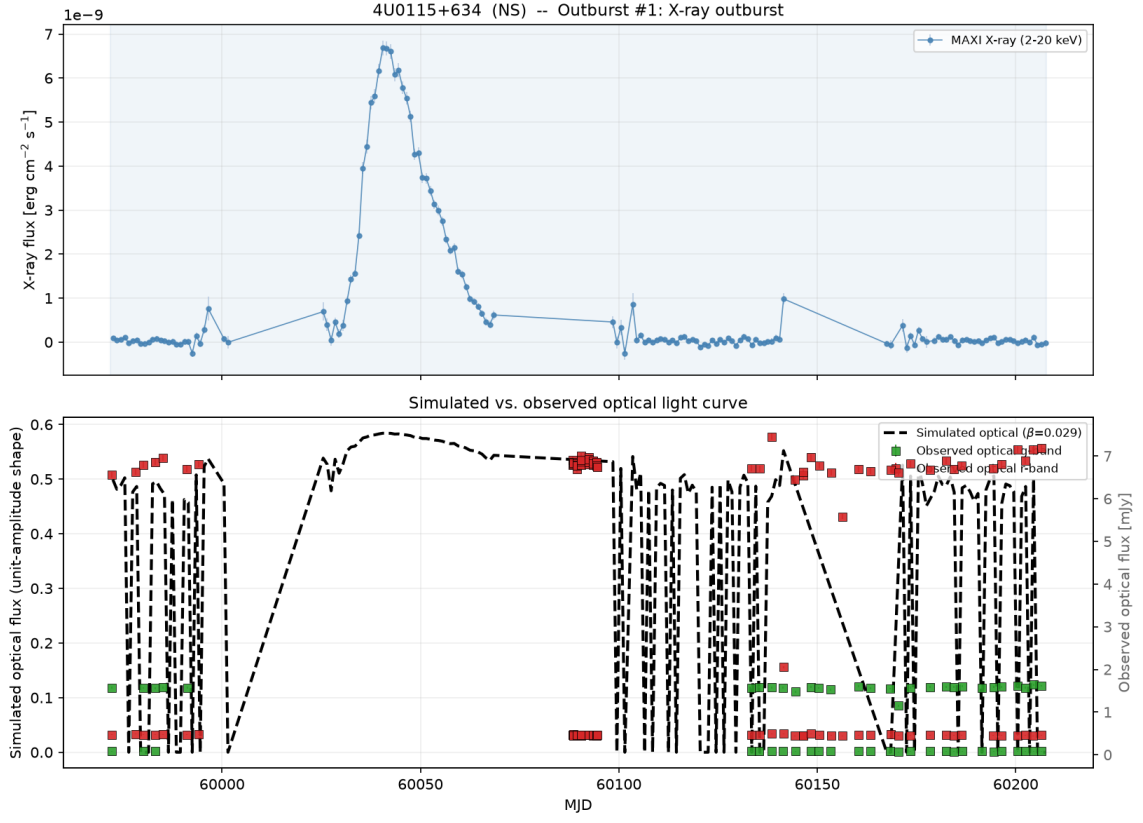


Figure 13: 4U 0115+634 (NS), outburst #1 ($\beta_{\text{NS}} = 0.029$). Low-level X-ray noise produces a sawtooth optical artifact.

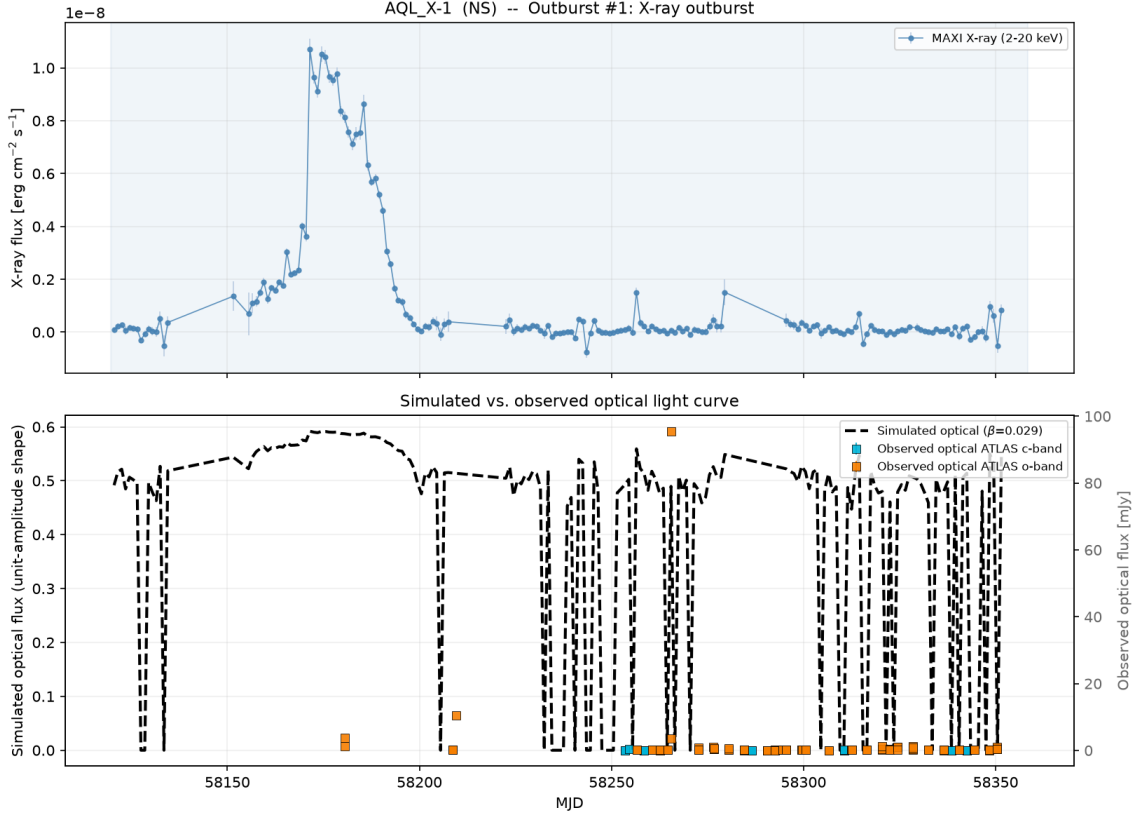


Figure 14: Aql X-1 (NS), outburst #1. The near-zero exponent causes unphysical optical spiking both during and outside the outburst.

This flattened slope stems from system-type contamination. The NS fit combined both LMXBs and HMXBs. In HMXBs like EXO 2030+375 (Figure 15), optical light is dominated by the luminous donor star rather than disc reprocessing, remaining virtually flat ($g \sim 0.018$ mJy) despite a major X-ray outburst. Pooling these flat HMXB curves dragged the overall NS reprocessing slope down toward zero.

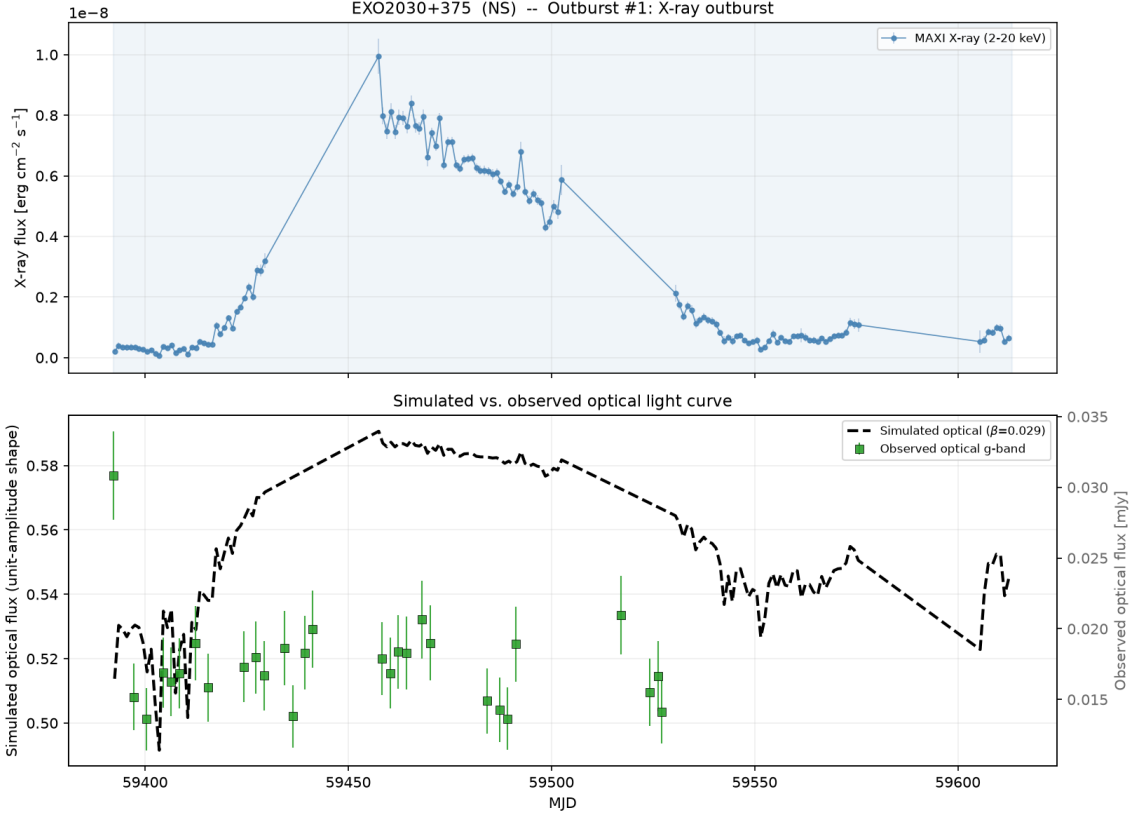


Figure 15: EXO 2030+375 (NS-HMXB), outburst #1. Optical flux remains flat during a massive X-ray flare, confirming donor-star dominance.

8.4 Key Pipeline Improvements

To address these findings, future model iterations should incorporate:

1. **A Minimum Slope Floor:** Implement a lower bound (e.g., $|\beta| \geq 0.1$) alongside the upper ceiling to filter out unphysical, flat fits.
2. **System-Type Stratification:** Separate LMXB and HMXB sources prior to fitting so donor-dominated optical emission does not contaminate disc-reprocessing estimates.
3. **Time-Lag Parameters:** Add an explicit delay parameter to capture the observed weeks-to-months lags between X-ray and optical peaks.
4. **Explicit Fallback Labels:** Clearly distinguish fitted values from default fallbacks in output plots and data tables.

9 Conclusion

Through this project, I designed and built an automated, end-to-end multiwavelength pipeline to process, classify, and model X-ray binary outbursts from archival data. Completing this work provided several key insights into both the physical mechanisms of compact objects and the practical realities of archival data analysis:

- **Data Pipeline Engineering & Catalog Matching:** I gained hands-on experience building multi-survey cross-matching pipelines across MAXI, Swift, ZTF, and ATLAS. Implementing a hierarchical resolution tree using cached local databases (WATCHDOG, BlackCAT, XRBcats, HEASARC) proved essential for processing hundreds of targets reliably while avoiding rate limits.
- **Outburst Population Dynamics:** Fitting 571 outbursts across 125 sources using a Norris pulse convolved with a top-hat kernel successfully recovered clear population differences. I observed

that neutron stars exhibit faster rise times than black holes, and that HMXBs form a compact, short-duration group compared to the much broader duration spread of LMXBs. Recovering these physical trends automatically verified the validity of my fitting pipeline.

- **Modeling Complex Physics & Edge Cases:** Building the optical reprocessing simulator highlighted the importance of testing simple empirical models against real-world noise. I learned how unaccounted physical factors—such as donor-star light in HMXBs flattening regression slopes or viscous propagation delays causing optical lags—can lead to model breakdown if not properly stratified.

Ultimately, this project gave me a deep, practical understanding of multiwavelength astronomy, light-curve modeling, and diagnostic pipeline design, establishing a clear foundation for future model refinements.

Looking ahead, the volume of reliably cross-matched X-ray and optical datasets will continue to grow as new time domain surveys come online. Scaling this pipeline to ingest larger datasets will enable stronger constraints on outburst evolution and optical reprocessing across different X-ray binary populations. The modular and automated framework developed in this project is therefore well positioned to support the Rubin LSST era, where millions of optical transients will be monitored over a decade, and can be readily extended to incorporate observations from the upcoming Roman Space Telescope. Together, these facilities will provide opportunities for large-scale multiwavelength transient studies, allowing this pipeline to evolve into a powerful tool for future X-ray binary population analysis.

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